

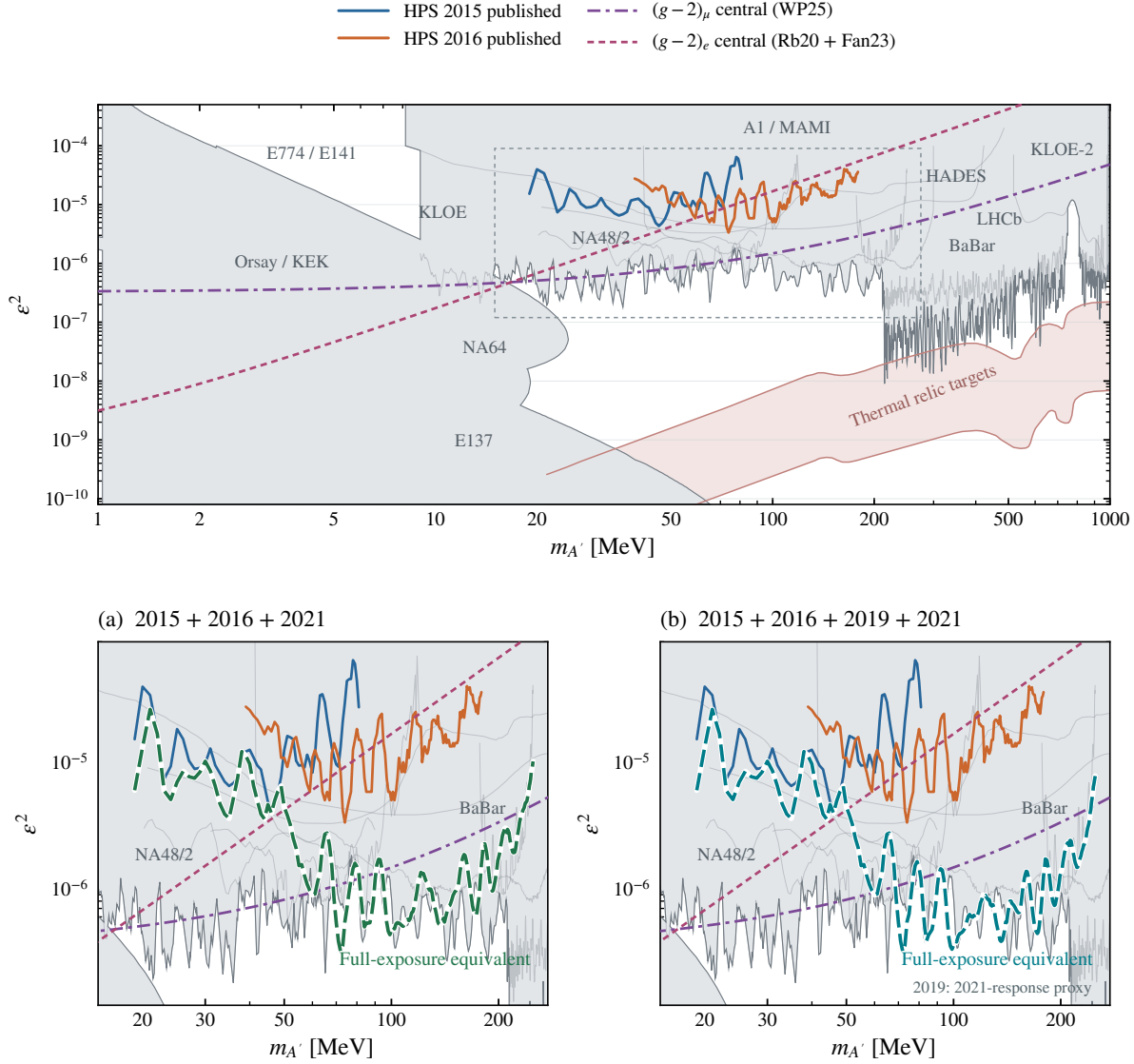


Figure 2: Visible dark-photon phase space and full-exposure-equivalent HPS projections. Top: the published engineering-run limits in their world-data context; the dashed rectangle marks the common zoom window in (a) and (b). Gray shading is the union of the archived visible exclusions, with their original boundaries retained. The world-data contours include BaBar, KLOE/KLOE-2, A1/MAMI, APEX, NA48/2, HADES, PHENIX, NA64, LHCb, and the older beam-dump or low-mass constraints from E137, E141, E774, Orsay, KEK, and U70 [5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21]. The thermal relic target band is shown as a benchmark for sub-GeV dark matter motivation [2]. The HPS 2015 and 2016 engineering-run prompt limits are highlighted using the legacy display convention that multiplies the published 95% ϵ^2 contours by 1.64/1.96 [3, 4]; the NA48/2 and BaBar inputs are treated in the same way. This approximate Gaussian-ratio rescaling is retained only for historical context and is not a recalculation of a 90% CL upper limit with the CL_s construction. The lepton central-value lines use $\Delta a_\mu = 38 \times 10^{-11}$ (WP25) and $\Delta a_e \simeq 0.34 \times 10^{-12}$ (Rb20 with Fan23) [22, 23, 24]. These are reference loci, not confidence bands; WP25 is compatible with zero. (a) Full-equivalent 2015+2016+2021; (b) full-equivalent 2015+2016+2019+2021. The latter adds the measured 2019 nominal 1% spectrum as a separate likelihood input with a 2021-response proxy. Both current-sample combined limits are then scaled by the square root of the current-to-full summed count-density ratio, taking 2021 from 10% to 100% and, in (b), 2019 from 1% to 100%. These are conditional observed-equivalent projections, not expected sensitivities or full-data exclusions; Appendix L gives the construction.